

Rowan Yu

Senior Unreal Engine Software Engineer | Developer Enablement (SDK Plugins, Samples, Tutorials) | XR & Real-Time Performance

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PROFESSIONAL SUMMARY

Senior Unreal Engine Software Engineer with 8+ years of C++/Unreal experience across VR/XR simulation, multiplayer systems and developer-facing SDK work. At HTC VIVE R&D, built ViveOpenXR and SRWorks Unreal plugins, samples and tutorials, and supported external developers directly: questions and feature requests from HTC's developer community were routed to me, and I resolved them by debugging developers' integration issues and shipping plugin/sample updates driven by their feedback. Currently developing large-scale multiplayer military VR training simulators supporting up to 30 connected machines at Big X Reality. Native Mandarin speaker with working English proficiency. Recently completed a hands-on Unreal Android build validation (APK packaging, ADB deployment to an Android-based standalone headset, Logcat debugging) and currently ramping up on Android performance tooling such as Perfetto and Android GPU Inspector.

CORE SKILLS

Engines & Languages: Unreal Engine 4/5, C++, Blueprint, Unity, C#

Developer Enablement: SDK/plugin architecture (ViveOpenXR, SRWorks), developer samples, technical tutorials, developer community support, feedback-driven plugin/sample updates, pull request to EpicGames/UnrealEngine

Android: Unreal Android packaging, APK deployment via ADB, Logcat debugging, Android SDK/NDK workflow; ramping up on Perfetto and Android GPU Inspector

Multiplayer & Systems: Replication, RPC, client/server architecture, cross-client synchronization, PC-to-VR streaming, shared-memory IO

Performance: Unreal Insights, CPU/GPU profiling, async processing, multithreading, NPC optimization, runtime tuning

XR / VR: OpenXR, SteamVR, VIVE Pro 2, VIVE Focus 3, VIVE Tracker, SRWorks

Hardware & Data Integration: Arduino, Antilatency, OptiTrack, motion platforms, HTTP, Socket.IO, TCP/UDP

Collaboration & Community: Mentoring junior engineers, ad-hoc C++/Blueprint review, cross-functional debugging, regular Game Jam participant

Languages: Mandarin Chinese (native), English (working proficiency)

PROFESSIONAL EXPERIENCE

Big X Reality - Senior Unreal Engineer | 2022/08 - Present | Taiwan

- Developed high-fidelity multiplayer military VR training simulators for Windows PC and PC-to-VR streamed clients, with performance-sensitive rendering and client/server synchronization.
- Scaled multiplayer training scenarios to 30 connected computers mixing desktop clients and PC-VR streamed headsets.
- Completed an end-to-end Unreal Android build validation: packaged an APK, deployed it to an Android-based standalone XR headset (VIVE Focus 3) via ADB, launched it on device and inspected runtime behavior with Logcat.
- Diagnosed frame-rate drops in packaged builds using Unreal Insights, tracing bottlenecks to navigation-mesh tile updates and NPC workloads, and applied targeted optimizations.
- Designed reusable Unreal plugins and components for NPC behavior, NPC performance optimization, data collection and project-wide spawn/generation workflows; the NPC optimization component was designed to port cleanly to any Unreal project.
- Created a data collection abstraction layer so teammates could attach telemetry capture at feature level, while the system handled encryption, timestamp conversion and delivery to backend services through HTTP/Socket.IO.
- Co-developed motion-platform and Antilatency integration plugins using shared-memory IO for real-time platform control; integrated Arduino-based weapon peripherals and OptiTrack external tracking.
- Built .bat-based launch tooling to start the server and selected client types in one step, cutting repetitive multiplayer packaged-build test setup for the team.

- Built external Excel/config-driven runtime tuning tools so non-Unreal stakeholders could adjust packaged builds live during customer validation sessions.
- Mentored 2 junior engineers on Unreal development, Perforce, Jenkins and Git workflows; provided ad-hoc C++/Blueprint reviews.

HTC Corporation - Engineer, VIVE R&D | 06/2019 - 08/2022 | New Taipei City, Taiwan

- Supported external Unreal developers through HTC's developer community: developer questions and feature requests reported in community channels were assigned to me, and I troubleshoot integration issues, answered developers directly and shipped ViveOpenXR / SRWorks plugin and sample updates based on their feedback.
- Developed the ViveOpenXR Plugin for UE4.27 and UE5.0, covering OpenXR Scene Understanding, OpenXR Facial Tracking and VIVE Cosmos controller mapping; the UE5.0 plugin was submitted as a pull request to the EpicGames/UnrealEngine GitHub repository.
- Ported SRWorks Scene Understanding low-level C++ APIs into the OpenXR architecture, adapting API behavior to comply with OpenXR specification rules.
- Implemented the OpenXR Facial Tracking module using Unreal Live Link to stream facial expression data into engine animation workflows.
- Created developer-facing samples (Facial Tracking with MetaHuman avatars, Scene Understanding with OpenXRARTrackedGeometry, Hand Tracking, Eye Gaze dartboard demo) and authored tutorials so developers could adopt HTC-supported OpenXR features faster.
- Built SRWorks UE4 Experience and released SRWorks Runtime (.msi installer), Unreal Plugin and Samples to the developer community.

Artlord Studio - Engineer, R&D | 07/2016 - 12/2018 | Kaohsiung, Taiwan

- Built multiplayer VR shooting game systems using UE4 Replication and VR IK, applying profiling to reduce network traffic for commercial location-based deployment.
- Integrated HTC VIVE, Kinect, Fingo hand tracking and a single-axis motion platform across multiple location-based VR entertainment projects.

COMMUNITY

- Regular participant in Game Jam events, collaborating with new teammates under tight time constraints to prototype and ship playable builds.

EDUCATION

Southern Taiwan University of Science and Technology - B.S. Multimedia and Entertainment Science